Sound Quality Required	Bandwidth	Sampling rate	Number of bits	Data rate (bits/sec)	Comments
High fidelity music (compact disc)	5 Hz to 20 kHz	44.1 kHz	16 bit	706k	Satisfies even the most picky audiophile. Better than human hearing.
Telephone quality speech	200 Hz to 3.2 kHz	8 kHz	12 bit	96k	Good speech quality, but very poor for music.
(with companding)	200 Hz to 3.2 kHz	8 kHz	8 bit	64k	Nonlinear ADC reduces the data rate by 50%. A very common technique.
Speech encoded by Linear Predictive Coding	200 Hz to 3.2 kHz	8 kHz	12 bit	4k	DSP speech compression technique. Very low data rates, poor voice quality.

TABLE 22-2
Audio data rate vs. sound quality. The sound quality of a digitized audio signal depends on its *data rate*, the product of its sampling rate and number of bits per sample. This can be broken into three categories, high fidelity music (706 kbits/sec), telephone quality speech (64 kbits/sec), and compressed speech (4 kbits/sec).

discussed later in this chapter. An 8 kHz sampling rate, with an ADC precision of 8 bits per sample, results in a data rate of 64k bits/sec. This is the *brute force* data rate for natural sounding speech. Notice that speech requires less than 10% of the data rate of high fidelity music.

The data rate of 64k bits/sec represents the straightforward application of sampling and quantization theory to audio signals. Techniques for lowering the data rate further are based on *compressing* the data stream by removing the inherent redundancies in speech signals. Data compression is the topic of Chapter 27. One of the most efficient ways of compressing an audio signal is **Linear Predictive Coding (LPC)**, of which there are several variations and subgroups. Depending on the speech quality required, LPC can reduce the data rate to as little as 2-6k bits/sec. We will revisit LPC later in this chapter with *speech synthesis*.

High Fidelity Audio

Audiophiles demand the utmost sound quality, and all other factors are treated as secondary. If you had to describe the mindset in one word, it would be: *overkill*. Rather than just matching the abilities of the human ear, these systems are designed to *exceed* the limits of hearing. It's the only way to be sure that the reproduced music is pristine. Digital audio was brought to the world by the **compact laser disc**, or **CD**. This was a revolution in music; the sound quality of the CD system far exceeds older systems, such as records and tapes. DSP has been at the forefront of this technology.